

Status Competition under Income Inequality

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Status-related expenditures rise with income inequality, contrary to post-war predictions. We model a single high-income participant competing against Q low-income participants in probabilistic tournaments combining innate ability and acquired social capital, where social capital requires a threshold expenditure. Under diminishing wealth utility, exactly two stable Nash equilibria exist: universal and elite-only participation. A closed-form participation threshold decreases in competition size and in the social capital weight. Under reference-dependent preferences, a third equilibrium arises in which only low-income participants invest. Collective low-income mobility is permanently capped below one-half, with the binding ceiling tightening as the equilibrium converges toward elite-only participation. *Keywords:* status competition; income inequality; social capital; participation threshold; mobility ceiling

JEL classification code: [e.g. D31, D62, J62, Z13]¹

1 Introduction

The decision to invest in elite professional credentials is rarely reducible to expected returns on human capital alone. Research on elite hiring suggests that such investments also function as bids for social position: employers in top-tier professional firms consistently privilege the prestige of credentials over their content, and access to the most rewarding positions is shaped as much by social capital as by demonstrated skill (Rivera, 2015; Bills, 2003). Yet the relationship between income inequality and the propensity to make such investments remains poorly understood. Post-war scholars of status competition — Hirsch (1977), Galbraith (1958), and Frank (1985) — argued that rising living standards and narrowing income gaps intensify positional rivalry, as more participants can afford to compete for a fixed stock of social rank. Empirical research, on the other hand, often finds the opposite: status-related expenditure has risen with inequality, not with its reduction (Jin, Li, and Wu, 2011; Jaikumar and Sarin, 2015; Chai, Kaus, and Kiedaisch, 2019). This paper develops a theoretical model

¹See <https://www.aeaweb.org/econlit/jelCodes.php>.

that clarifies the conditions under which each pattern can arise.

The source of the tension, we argue, lies in the failure to endogenise status within an economic framework that takes participation costs seriously. Such models treat status competitions as fixed social facts rather than equilibrium outcomes, leaving the effect of inequality on participation indeterminate. A model that derives participation decisions endogenously — under income heterogeneity and a threshold cost of entry — can recover the empirical result as an equilibrium phenomenon and also identify the parametric conditions under which the post-war pattern might also hold.

We model a hierarchy of income positions in which one high-income participant competes against Q low-income participants through probabilistic tournaments combining innate ability and acquired social capital, with participants making intertemporal investment decisions over T periods. Social capital — the skills, credentials and network affiliations that institutional engagement makes available — can be obtained only by paying a threshold cost $\bar{v}$, making it accessible to the high-income participant by default and contingent on budget feasibility for low-income participants. The probability of advancement depends on both components, with social capital weighted more heavily than raw talent. Following Postlewaite (1998), status does not enter utility directly but operates through its effect on income and the intertemporal wealth utility it enables.

Four results emerge. First, under diminishing utility of wealth, only two stable Nash equilibria exist: one in which both high-income and low-income participants invest in social capital, and one in which only the high-income participant does. This binary structure formalizes the intuition behind Hirsch’s social limits to growth (Hirsch, 1977) and provides a microfoundation for the empirically observed stratification of status investments by income class. Second, the transition between these equilibria is governed jointly by the income gap $y_2 - y_1$ and the participation cost $\bar{v}$: wider inequality sustains participation incentives for the high-income participant even as it prices out low-income participants, while falling participation costs can restore universal competition. Taken together, these two forces explain why rising inequality and rising status expenditure can coexist. Third, we characterise precisely when the standard equilibrium structure breaks down. Under reference-dependent preferences satisfying $U_1 < U_2$, a third equilibrium exists in which only the low-income participants invest. We state the necessary and sufficient conditions for this equilibrium as a formal proposition and show that the feasibility window is governed by the probability surplus $\Phi(Q)$, which decreases in the weight placed on social capital. This result provides a formal account of how meritocratic beliefs and income segregation can reinforce one another (Mijs and Savage, 2020; Mijs, 2021). Fourth, while individual low-income mobility probability vanishes as $Q \rightarrow \infty$, collective class-level mobility converges to two distinct ceilings depending on which equilibrium prevails: 1 under universal participation and a value depending on relative importance of social capital under the elite-only participation. The elite-only participation ceiling is the binding one: the comparative statics on Q drive the equilibrium toward elite-only participation at exactly the rate collective mo-

bility approaches this hard limit, making it the operative social limit to growth in the empirically relevant regime.

In what follows, Section 2 surveys the literature on status, inequality and social capital investment, situating the present model against its closest quantitative predecessors. Section 3 develops the model, deriving winning probabilities for all four investment cases under general Q , characterising the Nash equilibria, deriving the participation threshold in closed form, and establishing the mobility ceilings. Section 4 discusses conclusions and policy implications.

2 Literature Survey

The central empirical puzzle this paper addresses is the divergence between the post-war theoretical literature and contemporary empirical evidence. Hirsch (1977), Galbraith (1958, 1994), and Frank (1985) argue that improved living standards and compressed income distributions intensify positional rivalry, as more participants can afford to compete for a fixed stock of social rank. Empirical research, however, consistently finds that status-related consumption rises with inequality rather than falling with it. Studies of education expenditure in China (Jin, Li, and Wu, 2011), conspicuous consumption in India (Jaikumar and Sarin, 2015), and broader cross-national data (Chai, Kaus, and Kiedaisch, 2019) all report this positive association. Mijs (2021) further documents how rising inequality has coincided with stronger meritocratic beliefs and increased segregation of status competitions, rather than heightened competition across class lines. Our model clarifies this divergence by identifying the equilibrium conditions under which the relationship between inequality and participation switches direction — a distinction the post-war literature could not make without endogenous participation.

The paper is motivated by the concerns about reconciliation of objective and subjective notions of status in the sociological literature (Wegener, 1992). Status considerations do complicate relationship between inequality and competitions. Although status or prestige has long been recognized as a key motivator of labor participation (Bénabou and Tirole, 2003; Frank, 1985), the conventional view contrasts status as a social goal against the economic goals of a consumer — a dichotomy that complicates the endogenisation of status within economic models. Yet status is neither independent of nor reducible to economic rewards (Goode, 1978; Moldovanu, Sela, and Shi, 2007). The sociological literature emphasizes the need to reconcile objective and subjective aspects of status (Wegener, 1992), highlighting how both structural position and perceived rank shape behavior. The competition-view of status that we adopt allows us to define explicitly the scope of income inequality’s effects by considering the motivations for objective status in a hierarchy of social positions. Within this framework, status influences economic outcomes indirectly: it does not enter utility directly through wealth, but through its effect on participation and relative ranking (Postlewaite, 1998).

The social capital literature provides the investment-theoretic foundation for the participation cost structure. Jencks (1979) highlights the dual economic and

social character of social capital, which mediates how inequality interacts with long-term investments in credentials and networks. Participation costs in elite competitions — tuition fees, membership dues, the opportunity cost of network cultivation — are threshold rather than continuous expenditures: they must be paid in full to unlock the social capital they provide (Heckman and Mosso, 2014). This threshold structure is central to the model’s ability to generate binary equilibrium outcomes and a closed-form participation condition.

The present paper constructs a model that combines the indirect status utility of Postlewaite (1998), the tournament structure of the contests literature, and the threshold cost structure from social capital theory into a single framework in which participation is an endogenous equilibrium outcome. The key architectural choice — fixing one high-income participant and varying the pool of low-income challengers Q — allows us to derive competition-size comparative statics as well as the divergence between individual and collective mobility with respect to the concentration of higher socio-economic levels.

The two closest quantitative predecessors each address the question of how inequality shapes status competition, but reach conclusions that diverge from those of the present paper. Frank (1984) treats status as a reciprocal phenomenon: workers accept wage penalties to occupy the top of the local rank distribution, so wider wage dispersion reduces competition intensity — narrowing inequality intensifies it. Cole, Mailath, and Postlewaite (1992) derive status concern endogenously from the marriage market and show directly that agents from an economy with a more compact income distribution save more (Cole, Mailath, and Postlewaite, 1992, p. 1103): again, narrowing inequality intensifies status competition. In each case the mechanism operates through the intensity of behavior within a fixed participant pool; whether agents can afford to participate is not asked. Making participation income-constrained — the single architectural change in the present paper — breaks this pattern: wider inequality simultaneously sustains high-income investment and prices out low-income participants, permitting a positive association between inequality and status expenditure that neither predecessor can generate.

3 Model

3.1 Basic Setup

We consider a status competition in which a single high-income participant competes against $Q \geq 1$ low-income participants ($Q + 1$ total). The high-income participant occupies the high-income position y_2 and seeks to maintain it; each low-income participant receives income $y_1 < y_2$ and seeks promotion. Two characteristics determine performance in the competition. First, *raw talent* $r_i \sim \text{Uniform}[0, 1]$, which cannot be affected by expenditure. Second, *social capital* $s_i \sim \text{Uniform}[0, 1]$, which can be enhanced only by paying a threshold cost $\bar{v}$. Both r_i and s_i are independently distributed and unknown to participant i before the competition. The participant with the highest composite score

$\mu r_i + (1 - \mu)s_i$ wins the high-income position, where $\mu \in (0, 1)$ weights raw talent relative to social capital.

Since all Q low-income participants share the same income y_1 and all face the same investment decision, we analyze the participation problem from the perspective of a representative low-income participant. The high-income participant's decision is analogous. A participant pays $\bar{\nu}$ to access social capital (setting $s_i > 0$) or abstains and saves instead (setting $s_i = 0$, relying on raw talent alone). Crucially, no participant knows r_i or s_i prior to deciding whether to pay $\bar{\nu}$, so investment decisions depend only on the income class of the participant.

Status enters utility only indirectly: the winner obtains disposable income y_2 in the following period, the loser retains y_1 . This treatment, following Postlewaite (1998), preserves a standard wealth-utility framework while endogenising the social dimension of the competition.

Focusing on conditions where signaling value is significant, we assume throughout that social capital is weighted more heavily than raw talent:

ASSUMPTION 1 $\mu < \frac{1}{2}$

The distribution of the composite score $x_i \equiv \mu r_i + (1 - \mu)s_i$ under Assumption 1 is trapezoidal (derived in Appendix 5), with CDF $F_X(x)$ that is piecewise quadratic on $[0, \mu]$, linear on $[\mu, 1 - \mu]$, and piecewise quadratic on $[1 - \mu, 1]$.

The four investment cases are as follows.

- **Case I** — High-income invests ($\nu_2 = \bar{\nu}$), low-income do not ($\nu_1 = 0$).
- **Case II** — Low-income invest ($\nu_1 = \bar{\nu}$), high-income does not ($\nu_2 = 0$).
- **Case III** — Neither invests ($\nu_1 = \nu_2 = 0$).
- **Case IV** — Both invest ($\nu_1 = \nu_2 = \bar{\nu}$).

3.2 Equilibrium Conditions

3.2.1 Winning Probabilities

The derivations are in Appendices 5 and 6. The per-low-income-participant winning probabilities are:

$$\begin{aligned}
 (1) \quad p_1 &= \frac{\mu}{2(1 - \mu)(Q + 2)} && \text{(Case I)} \\
 (2) \quad p'_1 &= \frac{6 - 7\mu}{6(1 - \mu)} && \text{(Case II, } Q = 1) \\
 (3) \quad p_1 &= \frac{1}{Q + 1} && \text{(Cases III and IV)}
 \end{aligned}$$

In Case I, the low-income participant competes on raw talent alone against the

high-income participant's full composite score — a structural disadvantage that drives p_1 strictly below the symmetric baseline $1/(Q + 1)$. In Case II the roles are reversed: the investing low-income participant has the full score advantage, pushing p'_1 strictly above the baseline. For Cases III and IV, symmetry gives $1/(Q + 1)$ directly. For general $Q \geq 2$, p'_1 is computable as a rational function of μ of degree Q (Appendix 6); all equilibrium results for Case II follow from the ordering result alone, without requiring a closed form for p'_1 .

The high-income participant's probability follows from $p_2 = 1 - Q p_1$ in each case.

The fundamental ordering that drives the Nash equilibrium analysis follows directly from Assumption 1:

$$(4) \quad p_1 < \frac{1}{Q + 1} < p'_1 \quad \text{for all } Q \geq 1, \mu < \frac{1}{2}$$

Figure 1 illustrates these probabilities. Panel (a) shows that p_1 (Case I) lies strictly below the baseline $1/(Q + 1)$ and decays rapidly with Q . Panel (b) shows p_2 (Case I, high-income participant) lying well above the baseline. Panel (c) shows p'_1 (Case II, $Q = 1$) strictly above the baseline throughout. Panel (d) is discussed in Section 3.2.5.

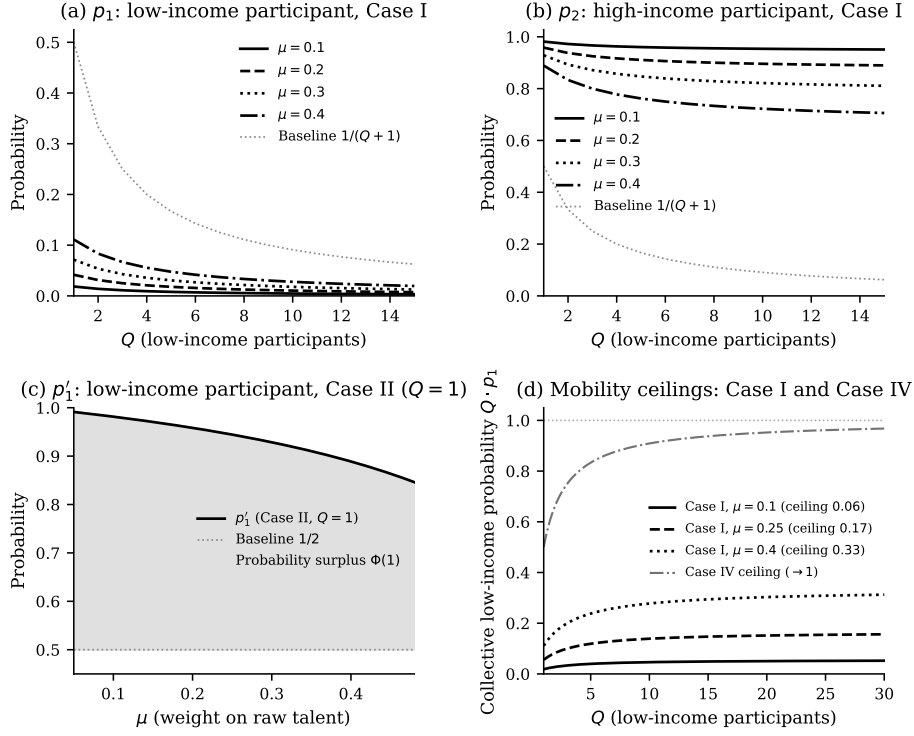


Figure 1

Winning probabilities and mobility ceiling. One high-income participant, Q low-income participants. Dotted horizontal lines in panels (a)–(b) show baselines $1/(Q+1)$ for each Q .

3.2.2 Intertemporal Choice and Utility Payoffs

Participants optimise an intertemporally additive utility $u(\cdot)$ over T periods. In period 1 each participant decides whether to pay $\bar{v}$; from period 2 onward she receives the income corresponding to her position. Define the T -period discount sum $D_T = \sum_{t=0}^{T-2} \delta^{-t} = (1 - \delta^{-(T-1)})/(1 - \delta^{-1})$, where $\delta > 1$ is the discount factor. The case $T = 2$ gives $D_T = 1$ and recovers the standard two-period model. The expected lifetime utilities are:

$$\text{Low-income participant: } u(y_1 - \nu_1) + \frac{D_T}{\delta} [p_1^c u(y_2) + (1 - p_1^c) u(y_1)]$$

$$\text{High-income participant: } u(y_2 - \nu_2) + \frac{D_T}{\delta} [p_2^c u(y_2) + (1 - p_2^c) u(y_1)]$$

where p_i^c denotes the winning probability in case $c \in \{\text{I, II, III, IV}\}$. Define the utility costs of participation:

$$U_1 \equiv u(y_1) - u(y_1 - \bar{v}), \quad U_2 \equiv u(y_2) - u(y_2 - \bar{v})$$

Under diminishing marginal utility with $y_1 < y_2$: $U_2 < U_1$. Table 1 summarises the four cases as a 2×2 game; full utility payoffs are in Table A1 (Appendix 8).

Table 1
Investment strategies

	<i>Low-income withdraws</i>	<i>Low-income invests</i>
<i>High-income withdraws</i>	Case III	Case II
<i>High-income invests</i>	Case I	Case IV

The full utility payoffs for each case are reported in Table A1 in Appendix 8; the equilibrium conditions below are derived from those payoffs.

3.2.3 Nash Equilibrium Conditions

For ease of notation define the probability-gap terms:

$$(5) \quad \begin{aligned} p_1 &\equiv \frac{\mu}{2(1-\mu)(Q+2)}, & p_2 &\equiv 1 - Q p_1 \\ p'_1 &\equiv \frac{6-7\mu}{6(1-\mu)} \quad (Q=1) \end{aligned}$$

The ordering $p_1 < 1/(Q+1) < p_2$ and $p'_1 > 1/(Q+1)$ holds for all $\mu < 1/2$ (Equation 4). Note that $p'_2 = 1 - Q p'_1$ satisfies $p'_2 < 1/(Q+1)$ by the same ordering, but p'_2 does not appear in the equilibrium conditions below: by the algebraic identity $1/(Q+1) - p'_2 = Q(p'_1 - 1/(Q+1))$, it is replaced throughout by $Q \cdot \Phi(Q)$, which is valid for all $Q \geq 1$.

Case I (high-income invests, low-income withdraws) is a Nash equilibrium if and only if:²

$$(6) \quad \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1)) < U_1$$

$$(7) \quad U_2 < \frac{D_T}{\delta} \left(p_2 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1))$$

Both conditions hold simultaneously under diminishing utility since $U_2 < U_1$ and $p_2 - 1/(Q+1) \geq 1/(Q+1) - p_1$. A higher $\bar{v}$ favors Case I so long

²Derived from the payoff table (Appendix 8): the low-income participant prefers not to invest, and the high-income participant prefers to invest, given the other's strategy.

as the high-income participant retains an incentive to invest; a lower $\bar{v}$ can destabilize it if low-income participants are induced to enter. Wider income differences ($y_2 - y_1$ larger) reinforce the high-income participant's incentive to invest while simultaneously raising the stakes enough to attract low-income investment, making the first condition harder to satisfy.

Case II (low-income invest, high-income withdraws). We now state this case as a formal proposition, since it requires conditions beyond standard diminishing utility and its characterisation carries independent theoretical content.

Define the probability surplus available to a low-income investor when the high-income participant withdraws:

$$(8) \quad \Phi(Q) \equiv \frac{D_T}{\delta} \left(p'_1 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1))$$

Since $p'_1 > 1/(Q+1)$ for all $\mu < 1/2$ (Equation 4), $\Phi(Q) > 0$ always. For $Q = 1$, substituting $p'_1 = (6 - 7\mu)/[6(1 - \mu)]$ yields:

$$(9) \quad \Phi(1) = \frac{D_T}{\delta} \cdot \frac{3 - 4\mu}{6(1 - \mu)} \cdot (u(y_2) - u(y_1))$$

where $(3 - 4\mu) > 0$ for all $\mu < 1/2$ under Assumption 1, so $\Phi(1)$ is strictly positive and decreasing in μ . For $Q \geq 2$, p'_1 is computable as a rational function of μ of degree Q for each specific Q , but no uniform closed-form expression holds across all Q simultaneously (as noted in Section 3.2). Nevertheless, $\Phi(Q)$ is computable numerically for any (Q, μ) and inherits the ordering $\Phi(Q) > 0$ for all $\mu < 1/2$. The qualitative conclusions of Proposition 1 — that Case II requires reference-dependent preferences for all $Q \geq 1$ and that the feasibility window $[U_1, U_2/Q]$ narrows as μ increases — hold for general Q by the ordering results in Equation 4.

PROPOSITION 1 (CASE II NASH EQUILIBRIUM) *Case II (low-income invest, high-income withdraws) is a Nash equilibrium if and only if:*

$$(10) \quad U_1 < \Phi(Q) \leq U_2/Q$$

A necessary condition is therefore $Q \cdot U_1 < U_2$. Under standard diminishing utility with $y_1 < y_2$, $U_2 < U_1$, so $U_2/Q < U_1$ for all $Q \geq 1$ and Condition (10) cannot be satisfied. Case II requires reference-dependent preferences under which $U_1 < U_2$ — the marginal utility cost of $\bar{v}$ is higher for the low-income participant than for the high-income participant, reversing the standard concavity ordering. For $Q = 1$, the condition reduces to:

$$(11) \quad U_1 < \Phi(1) \leq U_2$$

which is feasible if and only if $U_1 < U_2$.

PROOF Condition (10) is derived directly from the utility payoffs in Table A1

(Appendix 8). The low-income participant prefers to invest given the high-income participant withdraws if and only if $U_1 < \Phi(Q)$. The high-income participant prefers to withdraw given the low-income participants invest if and only if the gain from deviating to investment, $\frac{D_T}{\delta}(1/(Q+1) - p'_2)(u(y_2) - u(y_1)) = Q \cdot \Phi(Q)$ (using $1/(Q+1) - p'_2 = Q(p'_1 - 1/(Q+1))$, valid for all $Q \geq 1$), is less than the cost U_2 , i.e. $U_2 \geq Q \cdot \Phi(Q)$, equivalently $\Phi(Q) \leq U_2/Q$. Both conditions together require $U_1 < U_2/Q$. Under standard concavity $U_2 < U_1$, so $U_2/Q < U_1$ for all $Q \geq 1$ and the condition fails. $\square$

Several features of this proposition deserve comment. First, $\Phi(1)$ is strictly decreasing in μ : as social capital is weighted more heavily, the probability surplus from being the sole investor narrows, and the Case II window must be correspondingly tighter to sustain the equilibrium. At $\mu \rightarrow 0$ (pure raw-talent competition), $\Phi(1) \rightarrow 1/2$, giving the widest possible window; at $\mu \rightarrow 1/2$ (Assumption 1 binding), $\Phi(1) \rightarrow 1/3$. Second, the equilibrium remains fragile even when it exists: higher $\bar{v}$, greater impatience (δ rising), or wider income differences ($y_2 - y_1$ larger) can each collapse it into Case III or Case IV respectively (Mijs and Savage, 2020; Mijs, 2021).

Case III (neither invests) is a Nash equilibrium if:³

$$\begin{aligned} U_1 &> \frac{D_T}{\delta} \left(p'_1 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1)) \\ U_2 &> \frac{D_T}{\delta} \left(p_2 - \frac{1}{Q+1} \right) (u(y_2) - u(y_1)) \end{aligned}$$

Since $p'_1 - 1/(Q+1) \leq p_2 - 1/(Q+1)$ and $U_2 < U_1$, the binding constraint is the second inequality. Case III prevails when participation costs are prohibitively high relative to the income gain. If there is any net utility from competition, Case IV is the likely outcome.

Case IV (both invest) is a Nash equilibrium if:⁴

$$(12) \quad U_1 < \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1))$$

$$(13) \quad U_2 < Q \cdot \Phi(Q)$$

With $U_2 < U_1$, this equilibrium rests on the low-income participant's incentive: so long as the low-income participant finds the competition worthwhile, the high-income participant also participates. Note that Equations 6 and 12 are exact complements — Case IV obtains precisely where Case I fails for the low-income participant. The high-income participant's condition $U_2 < Q \cdot \Phi(Q)$

³Both participants prefer withdrawal given the other withdraws.

⁴Both participants prefer investment given the other invests.

uses the algebraic identity $1/(Q+1) - p'_2 = Q(p'_1 - 1/(Q+1))$, which holds for all $Q \geq 1$ and connects Case IV directly to the probability surplus $\Phi(Q)$ from Proposition 1: the high-income participant's boundary between Cases II and IV is exactly $Q \cdot \Phi(Q)$.

These conditions confirm that under diminishing utility only Cases I and IV are generically stable. For very low $\bar{v}$, both participants invest (Case IV); for very high $\bar{v}$, only the high-income participant invests (Case I); for sufficiently extreme $\bar{v}$, neither invests (Case III). Case II requires a non-standard utility structure and remains fragile throughout.

3.2.4 Participation Threshold

The boundary between Cases IV and I is defined by the critical cost $\bar{v}^*$ at which the low-income participant is indifferent between investing and withdrawing, given the high-income participant invests. Setting Equation 12 as an equality:

$$(14) \quad u(y_1) - u(y_1 - \bar{v}^*) = \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1))$$

Under log utility $u(y) = \log(y)$, this yields $\bar{v}^* = y_1(1 - (y_1/y_2)^A)$, showing that the threshold depends on the income ratio y_1/y_2 rather than the income level alone; the full derivation is in Appendix 7.

Competition size (Q increasing, more low-income challengers):

$$(15) \quad \frac{\partial A}{\partial Q} = \frac{-\mu(Q+1)^2 + 2(1-\mu)(Q+2)^2}{2(Q+1)^2(Q+2)^2(1-\mu)} \cdot \frac{1}{\delta}$$

Since $\mu < 1$, the denominator $2(Q+1)^2(Q+2)^2(\mu-1) < 0$, while the numerator $N = 2(1-\mu)(Q+2)^2 - \mu(Q+1)^2$ satisfies $N > (Q+2)^2 - \frac{1}{2}(Q+1)^2 = \frac{1}{2}Q^2 + 3Q + \frac{7}{2} > 0$ for all $Q \geq 0$ under $\mu < \frac{1}{2}$. Hence $\partial A / \partial Q < 0$. As the pool of low-income challengers grows, A decreases and $\bar{v}^*$ falls. Equivalently, a larger Q makes Case I increasingly likely to prevail over Case IV. As $Q \rightarrow \infty$, $A \rightarrow 0$ and the participation window closes entirely.

Competition size and real-world competition structures. The comparative statics on Q and μ map onto an observable empirical distinction between small and large competitions.

In small competitions — partner-track interviews at professional firms, senior academic appointments, board-level selections — the field of challengers is narrow and assessors can observe social capital directly: cultural fit, network affiliation, institutional credentials and communication style are all legible at close range. This corresponds to low μ (social capital decisive) and low Q . The model implies that in this regime the participation window A is relatively wide, both Cases I and IV are plausible depending on cost levels, and the mobility

ceiling $\mu/[2(1 - \mu)]$ is already binding in proportional terms. Individual low-income participants have non-negligible probabilities, yet class-level mobility is sharply constrained by μ . Case II is also most plausible here: the fragile equilibrium in which only the low-income participants over-invest in credentials — consistent with the sociological observation that aspiring candidates from lower-income backgrounds disproportionately invest in elite signals in highly selective, socially-embedded competitions (Heckman and Mosso, 2014; Mijs, 2021).

In large competitions — national civil service examinations, mass graduate recruitment schemes, standardised professional qualifications — the field is wide and assessors cannot observe social capital individually. Standardisation is typically designed to increase the weight on raw talent, i.e. to raise μ . The model implies two simultaneous effects. First, $\partial A/\partial Q < 0$ means the participation window closes as the field grows: the range of costs supporting universal participation shrinks, pushing toward elite-only equilibria regardless of intent. Second, raising μ through standardised assessment increases the ceiling $\mu/[2(1 - \mu)]$ and can partially offset this window-closing effect. The model thus provides a formal basis for the policy intuition that the *design* of large competitions — what they test, how accessible they are — matters more for class-level mobility than the design of small ones: in large competitions, the μ -effect on the ceiling and the Q -effect on the participation window are the two levers, and they must be managed jointly.

3.2.5 Collective Low-Income Probability and the Mobility Ceiling

The analysis so far has focused on the per-low-income-participant winning probability p_1 . A distinct and complementary object is the *collective low-income probability* $Q \cdot p_1$, which measures the probability that any low-income participant displaces the high-income participant — a statement about class-level mobility rather than individual opportunity. Two ceilings emerge depending on which equilibrium prevails.

Case I ceiling. Under Case I, the collective low-income probability in closed form is:

$$(16) \quad Q \cdot p_1 = \frac{Q\mu}{2(1 - \mu)(Q + 2)}$$

As $Q \rightarrow \infty$:

$$(17) \quad \lim_{Q \rightarrow \infty} Q \cdot p_1 = \frac{\mu}{2(1 - \mu)}$$

This ceiling is bounded strictly below $\frac{1}{2}$ for all $\mu < \frac{1}{2}$ and is increasing in μ .

Case IV ceiling. Under Case IV (both invest), all $Q + 1$ participants draw from the same distribution F_X , so $p_1 = 1/(Q + 1)$ and:

$$(18) \quad \lim_{Q \rightarrow \infty} Q \cdot \frac{1}{Q+1} = 1$$

Under universal participation, collective low-income mobility approaches certainty as the field grows.

Why the Case I ceiling is the economically relevant one. The two ceilings appear to give contradictory messages about the social limits to growth. The resolution lies in the comparative statics: by Equation 15, $\partial A / \partial Q < 0$, so a larger pool of low-income challengers *tightens* the participation window and makes Case I increasingly likely to prevail over Case IV. The Case I ceiling is therefore precisely the one that binds as Q grows: the equilibrium converges toward elite-only participation at exactly the same rate as collective mobility approaches its hard limit. The Case IV ceiling ($\rightarrow 1$) is relevant only when Q is small and costs are low enough to sustain universal participation — the regime where mobility concerns are least acute. It is the Case I ceiling $\mu/[2(1-\mu)]$ that constitutes the binding social limit to growth in the sense of Hirsch (1977).

Simultaneously, the *individual* low-income participant's probability $p_1 \rightarrow 0$ as $Q \rightarrow \infty$: each low-income participant's chances vanish even as collective mobility converges to the ceiling. This divergence between individual and collective mobility — the many trying, the class ceiling binding — is the formal content of the social limits argument.

4 Conclusion

This paper develops a formal model of status competition to clarify the conditions under which a persistent empirical tension arises. Post-war scholars predicted that narrowing income gaps would intensify positional rivalry; empirical research consistently finds the opposite. We show that this tension is tractable once participation in status competitions is treated as an endogenous equilibrium outcome shaped by the joint configuration of income differences and participation costs — and that the empirical pattern is the generic case under our model's assumptions.

The model fixes one high-income participant against a variable pool of Q low-income challengers competing through probabilistic tournaments that combine innate ability and acquired social capital. By deriving participation decisions from intertemporal utility maximisation over T periods, we obtain four principal results. First, under diminishing wealth utility, only two stable Nash equilibria exist: universal participation, in which both high-income and low-income participants invest in social capital, and elite-only participation, in which only the high-income participant does. This binary structure formalizes Hirsch's (1977) social limits to growth as an equilibrium theorem rather than a qualitative conjecture. Second, the transition between these equilibria is governed by a participation threshold $\bar{\nu}^*$ decreasing in both competition intensity Q and credential

weight μ . Wider inequality sustains high-income investment even as it prices out low-income participants, explaining why status expenditure and inequality can rise together. Third, reference-dependent utility introduces a boundary equilibrium — characterized by a formal proposition — in which only the low-income participants invest, requiring $U_1 < U_2$ and a feasibility window that narrows as social capital is weighted more heavily. Fourth, individual low-income mobility probability vanishes as $Q \rightarrow \infty$, but collective class-level mobility converges to $\mu/[2(1 - \mu)]$ under Case I and to 1 under Case IV. The Case I ceiling is the binding one: the comparative statics on Q drive the equilibrium toward elite-only participation at exactly the rate that collective mobility approaches this hard limit. It is therefore the Case I ceiling — not the trivial Case IV limit — that constitutes the operative social limit to growth in the empirically relevant regime.

The results add nuance to the predictions from models using fixed status hierarchies. Higher wage inequality in Frank (1984) reduces competition intensity as it lowers the cost of low status. In Cole, Mailath, and Postlewaite (1992), agents from an economy with a more compact income distribution save more, so narrowing inequality must raise status expenditures in both. The present model makes participation income-constrained and demonstrates how wider inequality simultaneously sustains high-income investment and prices out low-income participants — allowing a positive association between inequality and status expenditure.

For the analysis of inequality and status competition, the results clarify that reducing income differences without simultaneously lowering participation costs may leave elite-only equilibria intact. Conversely, policies that lower the threshold cost $\bar{v}$ — through subsidies to credential acquisition, reduced network barriers, or open-access professional qualifications — expand $\bar{v}^*$ and can restore the universal participation equilibrium. The mobility ceiling $\mu/[2(1 - \mu)]$ further implies that in competitions where institutional affiliation and credentials dominate raw ability, no amount of low-income participation alone can overcome the structural investment advantage of the high-income participant. Increasing μ — making raw talent more decisive relative to social capital — is therefore a necessary complement to cost-reduction policies if class-level mobility is to exceed the ceiling.

Several assumptions bound the analysis and point to natural extensions. The model restricts attention to a single high-income participant; relaxing this to multiple high-income participants preserves the binary equilibrium structure but changes the mobility ceiling formula, and is left to future work. The threshold cost $\bar{v}$ is treated as a recurrent period cost — consistent with ongoing credential maintenance, annual professional fees, and sustained network participation — rather than a one-time investment that low-income participants could save toward over multiple periods. A model with lumpy one-time investments and a savings margin would allow low-income participants to delay entry, potentially widening the feasibility window for Case IV; we leave this extension to future work. The threshold cost structure is binary — pay $\bar{v}$ in full or not at all

— which generates the clean closed-form results but abstracts from continuous investment intensities; a convex cost function would smooth the participation boundary and generate richer comparative statics on the income distribution. Most substantively, the incomes y_1 and y_2 are exogenous: the model identifies how inequality shapes participation given a distribution of positions, but not how status competition feeds back into the income distribution over time. Integrating the participation mechanism with an endogenous income process — closer in spirit to Cole, Mailath, and Postlewaite (1992) — would allow the model to speak to the long-run co-evolution of inequality and credential investment, which the empirical literature increasingly documents (Mijs, 2021; Mijs and Savage, 2020).

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5 Convolution of Uniform Random Variables

The distribution of a convex combination $w = \mu r + (1 - \mu)s$ of two random variables r, s - which are distributed according to densities $f_R(r)$ and $f_S(s)$ - can be derived using the convolution of two derived variables $x = \mu r$ and $y = (1 - \mu)s$. If r and s are uniformly distributed variables ranging from 0 to 1, then x ranges from 0 to μ and y ranges from 0 to $(1 - \mu)$. The CDF for $F_X(x)$ and $F_Y(y)$ are simply $F_X(x) = \frac{x}{\mu}$, $F_Y(y) = \frac{y}{1-\mu}$ (respectively). Therefore, the densities for x and y are $f_X(x) = \frac{1}{\mu}$ and $f_Y(y) = \frac{1}{1-\mu}$. The density of w is simple $f(w) = \int_{-\infty}^{\infty} f_Y(w - x)f_X(x)dx$. Since, $f_X(x)$ is $1/\mu$ only between 0 and μ . We have the following integral to evaluate for the convolution

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx$$

Since $f_Y(y)$ is 1 only between 0 and $1 - \mu$, we have the condition $0 < w - x < 1 - \mu \Rightarrow w - (1 - \mu) < x < w$. Further, x is distributed only between 0 and μ . As shown in Figure A1, the integral would evaluate to the following⁵ under the assumption $\mu < \frac{1}{2}$.

$$f(w) = \frac{1}{\mu} \int_0^{\mu} f_Y(w - x)dx = \begin{cases} \frac{1}{\mu} \int_0^w \frac{1}{1-\mu} dx & 0 < w < \mu \\ \frac{1}{\mu} \int_0^{\mu} \frac{1}{1-\mu} dx & \mu < w < 1 - \mu \\ \frac{1}{\mu} \int_{w-(1-\mu)}^{\mu} \frac{1}{1-\mu} dx & 1 - \mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

This can be simplified as

⁵Notice that as a convex combination, w is distributed only between 0 and 1.

$$f(w) = \begin{cases} \frac{w}{\mu(1-\mu)} & 0 < w < \mu \\ \frac{1}{1-\mu} & \mu < w < 1-\mu \\ \frac{1-w}{\mu(1-\mu)} & 1-\mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

Figure ?? shows this trapezoidal density with μ set to $\frac{1}{5}$. The CDF $F_X(x)$ can be evaluated by integrating the above density as $P(X < x) = \int_{-\infty}^x f(t)dt$.

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{\mu}{2(1-\mu)} + \frac{x-\mu}{1-\mu} & \mu < x < 1-\mu \\ \frac{\mu^2}{2\mu(1-\mu)} + \frac{2\mu(1-2\mu)}{2\mu(1-\mu)} + \frac{\mu^2-x^2+2x-1}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Simplifying above, we have

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

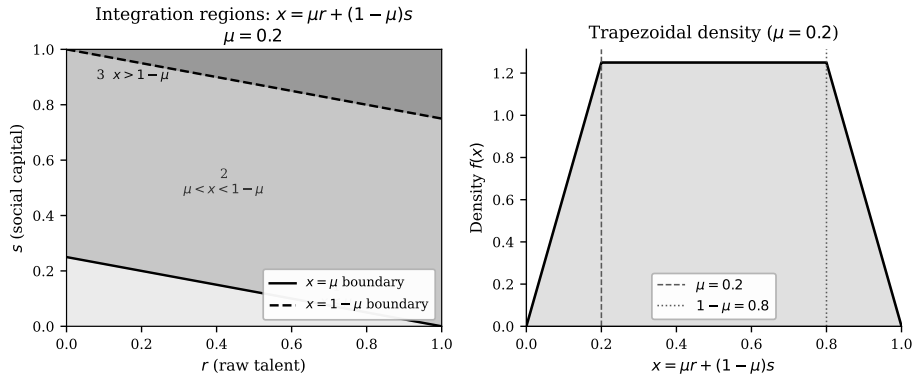


Figure A1

Left: ranges of w and x for convolution of two scaled random variables. Right: trapezoidal density of w as the convex combination with $\mu = \frac{1}{5}$.

6 Winning Probability Derivations

Case I: High-income invests, low-income do not

The low-income participant's score is μr_i (no social capital). She wins if μr_i exceeds all $Q - 1$ other low-income participants' scores and the high-income participant's full composite score $x_H = \mu r_H + (1 - \mu)s_H$. Since $\mu r_i < \mu$ for all $r_i \in (0, 1)$ under Assumption 1, only the first piece of F_X applies:

$$P_1(\text{win}|r_i) = r_i^{Q-1} \cdot F_X(\mu r_i) = r_i^{Q-1} \cdot \frac{\mu r_i^2}{2(1-\mu)} = \frac{\mu}{2(1-\mu)} r_i^{Q+1}$$

Integrating over $r_i \sim \text{Uniform}[0, 1]$:

$$p_1 = \int_0^1 \frac{\mu}{2(1-\mu)} r^{Q+1} dr = \frac{\mu}{2(1-\mu)(Q+2)}$$

Case II: Low-income invest, high-income does not

A given low-income participant wins if her full score $x_i = \mu r_i + (1 - \mu)s_i$ exceeds x_k for all $Q - 1$ other low-income participants and exceeds μr_H for the non-investing high-income participant. The conditional probability is:

$$P_1(\text{win}|r_i, s_i) = F_X(x_i)^{Q-1} \cdot G(x_i), \quad G(x) = \begin{cases} x/\mu & 0 < x < \mu \\ 1 & x \geq \mu \end{cases}$$

For $Q = 1$ the double integral over the unit square yields:

$$p'_1 = \frac{6 - 7\mu}{6(1-\mu)} \quad (Q = 1)$$

For $Q \geq 2$, $F_X(x)^{Q-1} \cdot G(x)$ produces a rational function of μ of degree Q within each CDF region; each specific Q yields a closed form but no compact general formula holds uniformly across all Q .

Cases III and IV

When neither invests, scores are μr_i ; by symmetry each of $Q + 1$ participants wins with probability $1/(Q + 1)$:

$$p_1 = \int_0^1 r^Q dr = \frac{1}{Q+1}$$

When both invest, all scores x_i are identically distributed via F_X ; by symmetry $p_1 = 1/(Q + 1)$ again.

Integration regions

To calculate $\iint P(\text{win}|r, s) dr ds$ over the unit square, we split by the boundaries of the piecewise function. The three regions are shown in Figure A2. If $x = \mu r + (1 - \mu)s$ and $F(x)$ is of the piecewise form with pieces A, B, C :

$$\iint F(x) ds dr = \int_{r=0}^1 \int_0^{\frac{\mu(1-r)}{1-\mu}} A ds dr + \int_{r=0}^1 \int_{\frac{\mu(1-r)}{1-\mu}}^{1-\frac{\mu r}{1-\mu}} B ds dr + \int_{r=0}^1 \int_{1-\frac{\mu r}{1-\mu}}^1 C ds dr$$

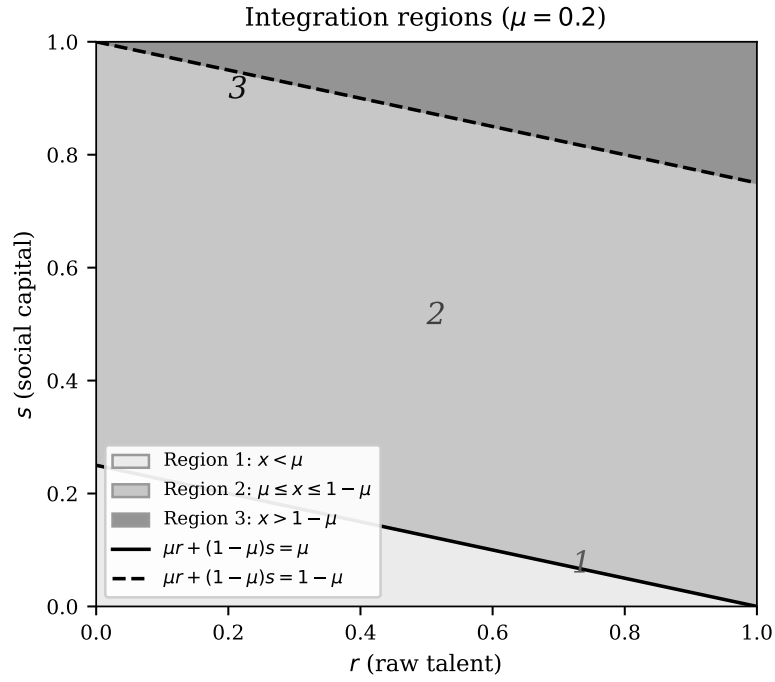


Figure A2

Integration over r, s split by boundaries $\mu r + (1 - \mu)s = \mu$ and $\mu r + (1 - \mu)s = 1 - \mu$

7 *T-Period Utility Generalisation*

The main text presents intertemporal utility under T periods with discount sum $D_T = \sum_{t=0}^{T-2} \delta^{-t}$. We establish here that the qualitative equilibrium structure is unchanged by T .

Closed form for D_T . Since $\delta > 1$, the geometric sum evaluates to:

$$D_T = \frac{1 - \delta^{-(T-1)}}{1 - \delta^{-1}}$$

For $T = 2$: $D_T = 1$, recovering the two-period model. For $T = 3$: $D_T = 1 + 1/\delta$.

Proposition (T-period invariance). *The set of Nash equilibria under T -period utility is identical to that under two-period utility ($T = 2$). The participation threshold $\bar{v}^*$ scales monotonically with D_T : it is strictly increasing in T and strictly decreasing in δ .*

Proof. All Nash equilibrium conditions (Equations 6–13) share the structure:

$$\frac{D_T}{\delta} \cdot \Delta p \cdot (u(y_2) - u(y_1)) \geq U_i$$

where Δp is a probability gap and U_i is a utility cost. Since D_T/δ is a common positive scaling factor, it does not alter which side of any inequality is larger: an equilibrium condition satisfied under $T = 2$ is satisfied for all $T \geq 2$ with the same qualitative direction. The threshold $\bar{v}^*$ (Equation A2) has $A \propto D_T/\delta$; since $(y_1/y_2)^A$ is decreasing in A and $y_1 < y_2$, $\bar{v}^* = y_1(1 - (y_1/y_2)^A)$ increases with A , hence with D_T . D_T is strictly increasing in T (adding positive terms) and strictly decreasing in δ (each term δ^{-t} is decreasing in δ). $\square$

The proposition implies that a longer investment horizon or greater patience raises $\bar{v}^*$, expanding the range of costs under which universal participation (Case IV) is sustained. The two-period model is therefore a conservative baseline: all comparative statics on $\bar{v}^*$ with respect to μ and Q carry over unchanged to the T -period case.

Log utility closed form

Under log utility $u(y) = \log(y)$, the participation threshold $\bar{v}^*$ has a tractable closed form. The indifference condition $U_1 = \frac{D_T}{\delta} \left(\frac{1}{Q+1} - p_1 \right) (u(y_2) - u(y_1))$ becomes:

$$(A1) \quad \log \frac{y_1}{y_1 - \bar{v}^*} = A \log \frac{y_2}{y_1}$$

where $A = \frac{D_T}{\delta} \left(\frac{1}{Q+1} - \frac{\mu}{2(1-\mu)(Q+2)} \right) > 0$. Solving:

$$(A2) \quad \bar{\nu}^* = y_1 \left(1 - \left(\frac{y_1}{y_2} \right)^A \right)$$

Since $(y_1/y_2)^A$ is decreasing in A and $y_1 < y_2$, $\bar{\nu}^*$ is increasing in A , hence decreasing in μ and Q (since $\partial A/\partial \mu < 0$ and $\partial A/\partial Q < 0$). The threshold depends on the income ratio y_1/y_2 rather than the income level alone.

8 Utility Payoffs

Table A1 reports the full utility payoffs for each of the four investment cases. In each cell, the upper entry is the high-income participant's utility and the lower entry is the low-income participant's utility. The scale factor D_T/δ multiplies all second-period terms.

Table A1
Utility payoffs (scale factor D_T/δ multiplies all second-period terms)

		<i>Low-income withdraws</i>	<i>Low-income invests</i>
<i>High-income withdraws</i>	High-income	$u(y_2) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Q u(y_1)}{Q+1} \right]$	$u(y_2) + \frac{D_T}{\delta} [(1 - Q p'_1) u(y_2) + Q p'_1 u(y_1)]$
	Low-income	$u(y_1) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Q u(y_1)}{Q+1} \right]$	$u(y_1 - \bar{\nu}) + \frac{D_T}{\delta} [p'_1 u(y_2) + (1 - p'_1) u(y_1)]$
<i>High-income invests</i>	High-income	$u(y_2 - \bar{\nu}) + \frac{D_T}{\delta} [p_2 u(y_2) + (1 - p_2) u(y_1)]$	$u(y_2 - \bar{\nu}) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Q u(y_1)}{Q+1} \right]$
	Low-income	$u(y_1) + \frac{D_T}{\delta} [p_1 u(y_2) + (1 - p_1) u(y_1)]$	$u(y_1 - \bar{\nu}) + \frac{D_T}{\delta} \left[\frac{u(y_2)}{Q+1} + \frac{Q u(y_1)}{Q+1} \right]$

p_1, p_2 : Case I probabilities. p'_1 : Case II low-income probability ($Q = 1$ closed form; general Q numerical). The high-income participant's Case II probability satisfies $1 - Q p'_1$, using the identity $1/(Q+1) - (1 - Q p'_1) = Q(p'_1 - 1/(Q+1))$, valid for all $Q \geq 1$.

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